Chapter 5 — Enzymes

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Enzymes act by: (a) Get consumed (b) Raise activation energy (c) Shift equilibrium (d) Lower activation energy
2. Protein part of an enzyme is: (a) Coenzyme (b) Prosthetic group (c) Holoenzyme (d) Apoenzyme
3. NAD+ is an example of a: (a) Substrate (b) Coenzyme (c) Apoenzyme (d) Prosthetic group
4. Lock-and-key model was proposed by: (a) Daniel Koshland (b) Louis Pasteur (c) Emil Fischer (d) Watson & Crick
5. In induced fit model, active site is: (a) Rigid (b) Made of DNA (c) Flexible, changes shape (d) Absent
6. A competitive inhibitor resembles the: (a) Cofactor (b) Product (c) Substrate (d) Enzyme protein
7. Penicillin inhibits: (a) Protein synthesis (b) Bacterial cell wall synthesis (c) Photosynthesis (d) DNA replication
8. Lyases catalyse: (a) Hydrolysis (b) Group transfer (c) Oxidation (d) Non-hydrolytic group addition/removal

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. How does the active site determine enzyme specificity?
2. Differentiate between cofactors and coenzymes.
3. How is an enzyme-substrate complex formed?
4. Differentiate between competitive and non-competitive inhibition.
5. How does feedback inhibition regulate metabolic pathways?
6. How are enzymes classified on the basis of reactions?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain the mechanism of enzyme action. **(4)**
(b) Compare the lock-and-key and induced fit models of enzyme action. **(4)**
Q.4 (a) Describe the types of enzyme inhibitors. **(4)**
(b) Classify enzymes on the basis of the reactions they catalyse. **(4)**

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